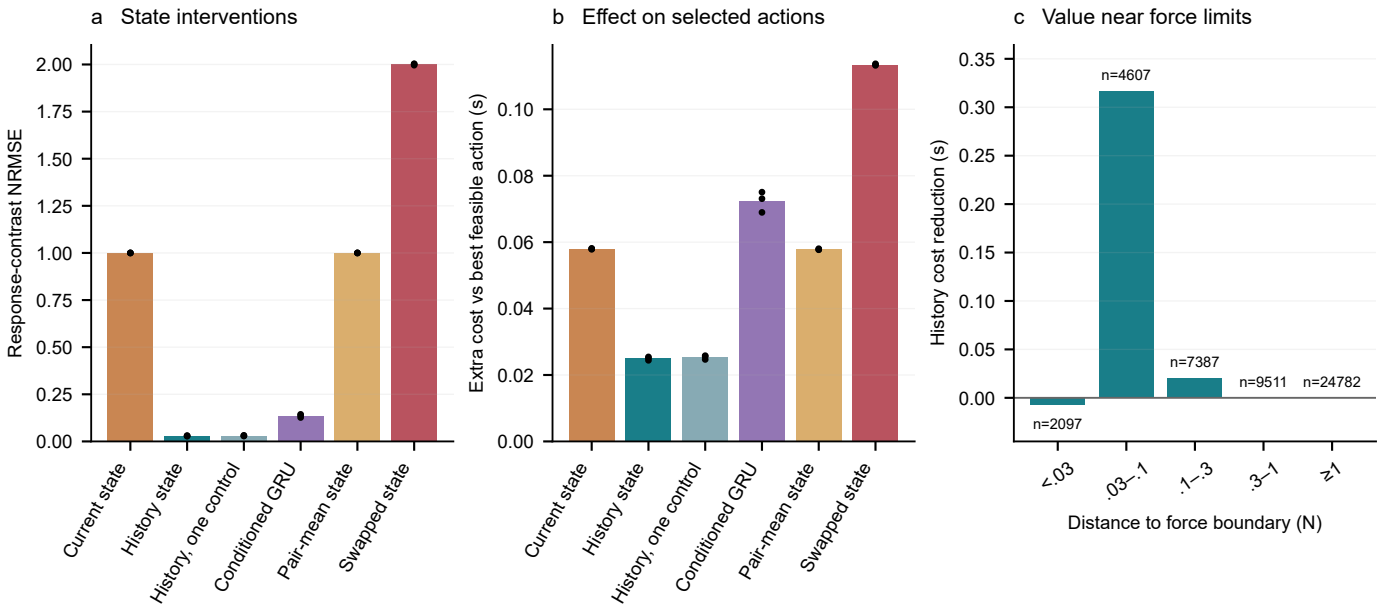
a State interventions
Response-contrast NRMSE
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Current state
History state
History, one control
Conditioned GRU
Pair-mean state
Swapped state
b Effect on selected actions
Extra cost vs best feasible action (s)
0.10
0.08
0.06
0.04
0.02
0.00
Current state
History state
History, one control
Conditioned GRU
Pair-mean state
Swapped state
c Value near force limits
History cost reduction (s)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
n=4607
n=7387
n=9511
n=24782
n=2097
<.03
.03–.1
.1–.3
.3–1
≥1
Distance to force boundary (N)